

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States both hypotheses using correct language	AO2.5	B1	$H_0: \lambda = 50$ $H_1: \lambda < 50$
	Selects and uses Poisson model with $\lambda = 50$ to find $P(\text{total vehicles} \leq 30)$ or $P(\text{total vehicles} < 30)$ or $P(\text{total vehicles} \leq 33)$ and $P(\text{total vehicles} \leq 34)$	AO3.3	M1	$X \sim \text{Po}(50)$ $P(X \leq 30) = 0.002$ $(P(X \leq 33) = 0.007 \text{ and } P(X \leq 34) = 0.0108)$ $p\text{-value} = 0.002 < 0.01$ $(30 < 34 = \text{critical value})$
	Obtains AWRP $P(X \leq 30) = 0.002$ (0.001594) or AWRP $P(X \leq 33) = 0.007$ (0.0069788) and AWRP $P(X \leq 34) = 0.0108$ (0.01078145)	AO1.1b	A1	Reject H_0 in favour of H_1 Significant evidence to suggest that the mean number of vehicles passing the service station per minute has reduced
	Evaluates the Poisson model by comparing 'their' p -value with 0.01 or 30 with their critical value	AO3.5a	R1	
	Infers H_0 rejected	AO2.2b	E1	
	Concludes in context (not definite)	AO3.2a	E1	
(b)	States meaning in context of Type I error	AO3.2a	E1	Type I error is to conclude that the mean number of vehicles passing the service station per minute using the junction has reduced when it has not.
(c)	States that rate at which events occur is unlikely to be constant over time in context or States that the events do not occur independently in context Must be consistent	AO3.5b	E1	The rate at which vehicles pass the service station is unlikely to be constant over time. or Vehicles may not pass the service station independently, eg convoy
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses Poisson $\lambda = 10$ PI	AO1.1a	M1	$V + W$ is $\text{Po}(10)$
	Obtains correct probability	AO1.1b	A1	$P(S > 10) = 1 - P(S \leq 10)$ $= 0.417$

(b)	States that model requires independence of purchases from store to store.	AO3.5b	E1	Purchases of printers at <i>Verigood</i> are not independent of those at <i>Winnerprint</i>
Total 3 marks				

Q3.

(a) (i) $e^{-1.5} \times 1.5^3 / 3!$

M1

$$= 0.126$$

0.125 to 0.126

A1

2

(ii) Using Po(1), $P(X < 1) = 1 - P(X \leq 1)$

M1

$$= 1 - 0.7358 = 0.264$$

SC Award M1 only if obtain 0.0902 using Po(0.5)

A1

2

(iii) Weekdays Po(7.5) weekend Po(1)
Weekdays = 7.5

M1

Total Po(8.5)

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A1

$$P(\text{Total} < 10) = P(\text{Total} \leq 9)$$

Applied (0.7764, 0.7166, 0.6530 are evidence)

m1

$$= 0.653$$

A1

4

(b) Using Total Po from (a)(iii)
 $P(> 15) = 0.0138, P(> 16) = 0.0066$

*M1 using their total providing supporting probabilities
seen OE use of $P(\text{Total} \leq 15 \text{ \& } 16)$*

M1

So needs 16 tubes

A1

2

- (c) Average rate of failure unlikely to be constant over the course of a day. Very little use of lights over this period.

One mark for any sensible comment

E1

1

[11]

Q4.

(a) (i)
$$\frac{e^{-3.5} \times 3.5^4}{4!}$$

M1

= 0.189

AWRT 0.189 Answer only gets B2

A1

2

- (ii) Using or stating Po(0.5)

An answer of 0.0144, 0.3935, 0.6065, 0.9098 or 0.9856 implies award of B1 but no further marks

B1

$P(\geq 2) = 1 - P(\leq 1)$

or

= 1 - 0.9098

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M1

= 0.0902

Accept 0.09

A1

3

- (iii) Using Po(14)

Sight of 0.1094, 0.1757, 0.9235, 0.9521

B1

$P(\leq 19) - P(\leq 10) = 0.9235 - 0.1757$

Allow 0.8752 - 0.1185

or 0.9573 - 0.2517 for M1

M1

$$= 0.7478$$

AWFW 0.747 to 0.748

A1

3

- (b) **GRBs / explosions / events / etc** will be random and / or independent

GRBs / etc short in comparison to observation period (non-overlapping)

For any valid point

E1

1

[9]

Q5.

- (a) (i) $X \sim P_o(06)$

$$P(X \leq 1) = 0.8781$$

Awrt 0.878

B1

1

- (ii) For matches:
The number of run outs: $Y \sim P_o(0.15)$

$$P(Y \geq 1) = 1 - P(Y = 0)$$

$$= 1 - e^{-0.15}$$

$$= 1 - 0.8607$$

must use $P_o(015)$.

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$$= 0.1393$$

awrt 0.139

A1

$$P(X \leq 1 \text{ and } Y \geq 1) = 0.8781 \times 0.1393$$

their (a)(i) $\times$ their $P(Y \geq 1)$

M1

$$= 0.122$$

awrt

A1

4

- (b) X and Y are independent.

Number of catches and run outs
independent

B1

1

(c) (i) For Season:

$$S \sim \text{Po}(9.6)$$

*Use of $\lambda = 9.6$ in correct
Poisson expression*

M1

$$P(S = 10) = \frac{e^{-9.6} \times 9.6^{10}}{10!}$$

$$= 0.124$$

A1

2

(ii) $T \sim \text{Po}(9.6 + 2.4) = \text{Po}(12)$
Po(12) used or seen

B1

$$\begin{aligned} P(T \geq 15) &= 1 - P(T \leq 14) \\ &= 1 - 0.7720 \\ &= 0.228 \\ (1 - 0.8444 &= 0.155 \text{ to } 0.156) \end{aligned}$$

B2,1

3

EXAM PAPERS PRACTICE [11]

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Q6.

(a) (i) $X \sim \text{Po}(13)$

Both Poisson and $\lambda = 13$

B1

1

(ii) $P(X = 20) = P(X \leq 20) - P(X \leq 19)$
 $= 0.975(0) - 0.957(3)$
[allow 0.975 – 0.957]

Must use $\lambda = 13$ otherwise MOA0

M1

$$= 0.0177 \text{ (3sf)}$$

AWFW 0.0176 to 0.018

$$\text{or } P(X = 20) = \frac{e^{-13} \times 13^{20}}{20!} \quad \text{M1}$$

$$\begin{aligned}
 \text{(iii)} \quad P(6 \leq X \leq 18) &= P(X \leq 18) - P(X \leq 5) \\
 &= 0.930(2)
 \end{aligned}$$

M1

$$- (0.0107 \text{ or } 0.0259)$$

M1

$$\begin{aligned}
 &= 0.920 \text{ (3sf)} \\
 &\text{AWFW } 0.919 \text{ to } 0.92
 \end{aligned}$$

A1

- (b) Cars not random
 Cars not independent
Allow (number of) cars not random / not independent
- Mean and Variance of cars different / not equal
B1 for any one of these 3 statements
*Must indicate a reference to **cars***

B1

Mean / Average / λ / 2.6
Correct comment about value of $\lambda \neq 2.6$

greater / less / smaller / different / variable / not constant /
 too small / too large

Any combination (one from each group):

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Any contextual reason that suggests a change in traffic flow, eg due to:
 rush hour / congestion / traffic jams / accidents / work traffic /
 school traffic / peak time

*eg mean greater due to rush hour, **or** λ smaller due to
 congestion, **or** 2.6 too small due to school traffic*

B1

- (c) $Y \sim \text{Bin}(20, 0.2)$
 $P(Y \geq 5) = 1 - P(Y \leq 4)$

or:

$$1 - \left(\begin{matrix} 0.01153 + 0.05765 + 0.13691 \\ +0.20536 + 0.21820 \end{matrix} \right)$$

$$= 1 - 0.6296$$

$$1 - 0.6296$$

(Allow $1 - 0.8042$ seen for M1)

M1

$$= 0.37(0) \text{ (3sf)}$$

AWFW 0.37 to 0.3704

A1

2

(d) X and Y independent

Any statement which indicates two / both events are independent

B1

$$p = 0.0177 \times 0.3704$$

[their (c)] $\times$ [their (a)(ii)]

M1

$$= 0.00656 \text{ (3sf)}$$

AWFW 0.0065 and 0.0067

A1

3

[13]

Q7.

$$X \sim \text{Po}(\lambda)$$

(a) $E(Z) = 4 \times 2.5 + 30 = \mathbf{40}$

CAO

B1

$$\text{Var}(Z) = 4^2 \times 2.5$$

Use of $V(aX) = a^2 V(X)$

Ignore '+30'

M1

$$= \mathbf{40} \{= E(Z)\}$$

CAO AG

A1

Note:

$$4 \times 2.5 + 30 = 4^2 \times 2.5 \Rightarrow \text{B1 M1 A0}$$

$$\text{plus value of 40 quoted} \Rightarrow \text{B1 M1 A1}$$

3

(b) No values less than 30 are possible

No odd values are possible

Only even values are possible

Only values of 30, 34, 38, etc, are possible

Or equivalent

B1

1

[4]

Q8.

- (a) (i) $X \sim \text{Po}(7)$

$$P(X \leq 5) = 0.301$$

AWFW 0.300 and 0.301

B1

1

(ii)
$$P(X = 7) = \frac{e^{-7} \times 7^7}{7!}$$

M1

$$= 0.149$$

A1

$$\begin{aligned}
 &P(X \leq 7) - P(X \leq 6) \\
 &= 0.5987 - 0.4497 \text{ (M1)} \\
 &= 0.149 \text{ (A1)}
 \end{aligned}$$

2

(iii) $0.65 \leq p \leq 0.66$

$$P(X \leq 9) - P(X \leq 4)$$

B3

$$0.72 \leq p \leq 0.73 \quad \text{or} \quad 0.52 \leq p \leq 0.53$$

$$\begin{aligned}
 &P(X \leq 10) - P(X \leq 4) \\
 &P(X \leq 9) - P(X \leq 5)
 \end{aligned}$$

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(B2)

$$0.60$$

$$P(X \leq 10) - P(X \leq 5)$$

(B1)

3

- (b) No. telephone calls received per hour
 $= Y \sim \text{Po}(0.875)$

B1

1

- (c) (i) Maximum number = 4

B1

1

(ii) $P(Y < 4) = P(Y = 0, 1, 2, 3)$

$$= e^{-0.875} \left(1 + \frac{7}{8} + \frac{49}{128} + \frac{343}{3072} \right)$$

$$= 0.4169(1 + 0.875 + 0.3828 + 0.1117)$$

$$= 0.987740443$$

Any correct expression (B2)
or AFWF 0.987 to 0.988

B2

$$P(Y \geq 4) = 1 - 0.9877$$

$$1 - (\text{their } P(Y < 4))$$

M1

$$= 0.0123$$

AWFW 0.0122 and 0.0123

A1

4

SC
 $P(Y \leq 4) = 0.997 \text{ to } 0.998$
or any correct expression

B2

$$P(Y > 4) = 0.002 \text{ to } 0.003 \quad \text{M1 A0}$$

- (d) λ probably not constant
The number of calls in any time interval
of 1 hour is likely to vary throughout the day.

'System Down'
 $\Rightarrow$ *not independent*

E1

1

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[13]

Q9.

- (a) (i) $X \sim \text{Po}(5.0)$
 $\Rightarrow P(X < 4) = P(X \leq 3)$
(0.440 to 0.441) for B1

$$= 0.265$$

CAO

B2

2

- (ii) $Y \sim \text{Po}(1.5)$
 $\Rightarrow P(Y = 4) = \frac{e^{-1.5} \times (1.5)^4}{4!}$

M1

$$= 0.0471$$

(0.047 to 0.0471)

A1

2

(b) (i) $T = X + Y \sim \text{Po}(6.5)$

B1

$$\Rightarrow P(T > 5) = 1 - P(T \leq 5)$$

(1 - 0.2237) **or** (1 - 0.5265)

B1

$$= 1 - 0.369$$

$$= 0.631$$

B1

3

(ii) $p = {}^8C_7 (0.631)^7 (0.369) + (0.631)^8$

ft on their p from (b)(i)
Either part attempted

M1ft

$$p = 0.11758 + 0.02513$$

(both parts correct)

A1ft

$$= 0.143$$

AWFW 1.142 to 0.143 (CAO)

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A1

3

(c) (i) Mean = 8

CAO

B1

$$\text{Variance} = s^2 = 16.9$$

(sample variance = 15.2)
(AWRT)

B1

2

(ii) Poisson not a good model for data

B1dep

$$\text{Mean} \neq \text{Variance}$$

B1

2

[14]

Q10.

- (a) (i) $X \sim \text{Po}(9.0) \Rightarrow$
standard deviation = 3

B1

1

- (ii) $P(6 < X < 12)$
 $= P(X \leq 11) - P(X \leq 6)$

M1

$$= 0.8030 - 0.2068$$

M1ft

$$= 0.5962$$

CAO

A1

3

- (b) (i) $T \sim \text{Po}(11.5)$
CAO

B1

1

- (ii) $P(T \leq 1) = P(T = 0) + P(T = 1)$
 $= e^{-11.5} + 11.5e^{-11.5}$
 $= 0.000127$

Use of $T = 0$ and 1

M1

Substitute correctly into formula

M1

AWFW 0.000126 and 0.00013

A1

3

- (c) $\bar{x} = 12.0$ and $s^2 = 19.3$ ($s = 4.40$)
 $\sigma^2 = 17.4$ ($\sigma = 4.17$)

B1

Mean and variance very different
dep on s^2 (or σ^2)

E1

$\Rightarrow \text{Po}(12.0)$ not a suitable model
(dep E1)

E1

3

[11]

Q11.

(a) (i) $P(A=4) = \frac{e^{-3.5} \times (3.5)^4}{4!} = 0.189$

M1A1

2

(ii) $P(B \leq 6) = 0.762$

B1

1

(iii) $T = A + B \sim \text{Po}(8.5)$

$P(T \text{ fewer than } 10) = P(T < 10)$
Use of $\text{Po}(8.5)$

M1

$= P(T \leq 9)$

$T \leq 9$ attempted

M1

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$= 0.653$

CAO

A1

3

(b) $X \sim B(5, 0.653)$

$X \sim B(5, \text{their } p)$

B1

$P(X \geq 4) = \binom{5}{4} (0.653)^4 (0.347) + (0.653)^5$

M1

$= 0.31547 + 0.11873$

$= 0.434$

On their p from (a)(iii)

A1ft
3

(c) (i) $\bar{x} = 9.2$

B1

$$s^2 = 9.29$$

$$\sigma^2 = 8.36$$

B1
2

- (ii) Mean and variance have similar values which suggests that Poisson distribution may be appropriate

B1ftB1ft
2

[13]

Q12.

(a) (i) $P(X = 2) = \frac{e^{-1.5} \times (1.5)^2}{2!} = 0.251$

M1A1
2

(ii) $p = (0.251)^3 = 0.0158$
on their p from (i)

M1A1ft
2

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(b) (i) $Y \sim P_o(9.0)$

B1
1

(ii) $P(Y \geq 12) = 1 - P(Y \leq 11)$
 $= 1 - 0.8030$

M1

$$= 0.197$$

A1
2

- (c) attacks patients:
 randomly (p constant)
mean of 1.5 $\Rightarrow$ p small (B1)
(unless very few patients)

B1

independently

B1

[9]

Q13.

- (a) For a 1-year period
The number of A grades $\sim \text{Po}(3)$

For a 5-year period
The number of A grades $\sim \text{Po}(15)$

B1

$P(\text{Total A-grades} > 18)$
 $= 1 - (\text{Total} \leq 18)$

M1

$$\begin{aligned} &= 1 - 0.8195 \\ &= 0.1805 \\ &= 0.181 \end{aligned}$$

AWFW 0.180 and 0.181

A1

3

- (b) (i) $X + Y \sim \text{Po}(10)$

B1

$$\begin{aligned} P\{X + Y \leq 14\} &= 0.917 \\ &\quad (0.916 \text{ to } 0.917) \end{aligned}$$

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M1A1

3

- (ii) X and Y are independent variables

E1

1

[7]

Q14.

- (a) (i) $P(3) = 0.9212 - 0.7834 = 0.138$
 $P(3) = P(3 \text{ or fewer}) - P(2 \text{ or fewer})$ or
use of correct formula

M1

$$0.138 \text{ (0.1375 to 0.1385)}$$

A1

2

- (ii) Poisson mean 2.4
Poisson, mean 2.4

B1

$$P(0) = 0.0907$$

0.0907 (0.0907 to 0.09075)

B1

2

- (iii) Poisson mean 12
Poisson mean 12

B1

$$P(20 \text{ or more}) = 1 - P(19 \text{ or fewer})$$

P(20 or more) = 1 - P(19 or fewer)

M1

$$= 1 - 0.9787$$

$$= 0.0213$$

A1

3

0.0213 (0.021 to 0.0214)
sc allow B2 for 0.0116 (0.011 to 0.0117)

- (b) (i) Poisson mean 1.8
 Standard deviation $\sqrt{1.8} = 1.34$
ft their mean

M1

1.34 (1.34 to 1.345)

A1

2

- (ii) Cannot distinguish between $2 - 1$ and $3 - 0$
Reason

E1

1

- (c) Mean not constant
Reason - generous

E1

1

[11]

Q15.

- (a) Poisson

Poisson

B1
1

(b) (i) $x = 3.69$ $s^2 = 7.06$, allow 6.52

B1

3.69 (3.68 – 3.7)

B1
2

7.06 (7.06 – 7.07) or 6.52 (6.51 – 0 6.53)

(ii) No, mean and variance very different

M1

Attempted comparison of mean and variance

A1
2

*No - must be clearly based on variance
e.g. disallow if compared with s.d. even if they call it variance*

[5]

Q16.

(a) 0.977

*Correct use of tables/formula
0.977(0.976 – 0.977)*

M1/A1
2

(b) Poisson mean 3

Mean 3

B1

$P(>4) = 1 - 8153$

Method, allow 1 – 3 or fewer

M1

$= 0.185$

0.185 (0.185 – 0.186)

A1
3

(c) Individuals do not enter independently

E2(1) independent

E2(1)

Q17.

- (a) (i) 0.231
0.23 – 0.232

B1

1

- (ii) 1 – 0.4695
*P(3 or more) = 1 – P(2 or fewer) or
correct method of calculation*

M1

= 0.5305
0.53 – 0.531

A1

2

- (b) Poisson mean 5×2.8
completely correct method

M1 M1

$P(<10) = 0.109$
0.109 – 0.11

A1

3

- (c) (i) $x = 1.3$
cao

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B1

$S^2 = 3.34$
3.34 – 3.35
allow 3.01 (3.005 – 3.015)

B1

2

- (ii) Mean and variance different

B1

1

- (iii) Mean unlikely to be constant over 24 hour period/not independent

B1

1

Q18.

- (a) (i) $P(2 \text{ or fewer}) = 0.8335$
 $0.8335(0.833 - 0.834)$

B1

- (ii) Poisson mean 2.8
Poisson mean 2×1.4

M1

$$P(4 \text{ or more}) = 1 - 0.6919$$

$$P(4 \text{ or more}) = 1 - P(3 \text{ or fewer})$$

Method

M1 M1

$$= 0.308$$

A1

5

$$0.308(0.3075 - 0.3085)$$

- (b) Poisson mean 2.4

M1

$$P(<4) = 0.7787$$

A1

2

$$0.779(0.778 - 0.779)$$

- (c) Poisson mean 3.8

Poisson mean $1.4 + 2.4$

M1

$$P(> 6) = 1 - 0.9091$$

$$P(> 6) = 1 - P(6 \text{ or fewer})$$

m1

$$= 0.0909$$

$$0.0909(0.908 - 0.091)$$

A1

3

- (d) Not independent/ mean may vary according to day of the week

E1

1

[11]

Examiner reports

Q1.

In part (a), the majority of students scored at least three marks. Most students stated the correct hypotheses with some students using μ instead of λ . A minority of students stated hypotheses for a goodness of fit test or used the sample value in their hypotheses. Most students calculated the Poisson probability correctly. The most common error was to calculate the probability of exactly 30 rather than 30 and less. A significant proportion of students found the critical region instead but some students incorrectly calculated the region or did not provide enough evidence that they had found the critical region. When comparing the Poisson probability with the significance level, some students made errors in order of magnitude, either mistaking 0.00159 as 1.59% or 1% as 0.1 or 0.001. Some students lost the final mark for being too definite in their final conclusion or not writing a conclusion in context.

In part (b), most students correctly identified Type I error but many failed to do so in context. Some students described Type II error instead or described the probability of Type I error. A small amount of students were not careful with the use of double negatives in their language.

The majority of students did not score the mark for part (c). Many students criticised the time that Xander took the sample but did not relate this to the Poisson model assumption of a constant rate over time. A significant proportion of students declared that the process was not random whilst giving descriptions in context of the events not being independent.

Q3.

This question was generally very well-answered, although a small proportion used incorrect values of λ and scored poorly. In the final part, some students merely used 'standard' Poisson phrases such as 'not independent' or 'mean not equal to variance', but most 'considered the context and produced a sensible explanation.'

Q4.

Part (a)(i) was generally well answered and, in parts (ii) and (iii), the value of λ was usually correctly calculated, but the boundary values to be used caused more difficulty. In part (b), answers such as "The days are independent" or "The satellites are independent" showed why it could not be assumed that "They are independent" referred to the GRBs. A not uncommon answer was "Because mean and variance are equal" which showed that students were quoting learnt answers, without any regard for the information given, or not given, in the question.

Q5.

Part (a)(i) was answered very well. However, in part (a)(ii), some candidates attempted to use $Po(1.5)$ or, because $Po(0.15)$ was not in the booklet provided, attempted to average the values found under $Po(0.10)$ and $Po(0.20)$. Neither of these approaches gained any credit. In part (b), it was expected that candidates would state either that 'X and Y are independent' or that 'the number of catches and run-outs are independent'. Although these were often stated, there were candidates who thought that stating 'they are independent' or that 'cricket matches are independent' would suffice. These statements were not sufficient to gain any credit.

Parts (c)(i) and (c)(ii) were usually very well answered. However, some candidates

seemed to have some difficulty interpreting statements such as 'at least 15'.

Q6.

In part (a)(i), candidates were expected not only to give the value of λ to be 13 but also to state that X followed a Poisson distribution; this was not always seen. Part (a)(ii) was usually done well either by using tables or by applying the correct formula. Unfortunately some candidates used $\lambda = 2.6$. In part (a)(iii), tables were often used correctly and accurately to obtain the correct answer of 0.920. Part (b) was not done at all well, with only the most able candidates gaining any credit. Although many thought that the value of λ would be different from 2.6, they could not give a convincing reason why this should be the case.

Comments such as 'It is not random' or 'Mean not equal to the variance' by themselves were not specific enough and consequently gained no credit. Part (c) was usually very well answered by the vast majority of candidates who used tables. Those who chose to use the formula approach usually either missed one of the terms out or made some error in their calculations. In part (d), the necessary assumption that was expected was ' X and Y are independent'. Comments such as 'They are independent' or 'The speeds are independent' gained no credit. The calculation of the required probability was usually attempted correctly.

Q7.

Most candidates provided a convincing answer in part (b). Also, it was very rare to see an incorrect answer to part (a) but even rarer to see a correct reason in part (b), where the most common answer was ' λ too large', which, needless to say, did not score the mark.

Q8.

The majority of candidates gained full marks for parts (a)(i) and (a)(ii). However, in part (a)(iii), although there were many correct answers seen, there were far too many candidates who could not interpret correctly the expression 'at least 5 but fewer than 10'.

In part (b), candidates were asked to write down the **distribution** of the number of telephone calls received each hour. Although the majority stated that $\lambda = 0.875$, they did not state that the distribution was Poisson and consequently lost a mark. Part (c)(ii) was answered correctly by, in the main, only the most able candidates. Some who correctly realised that $P(Y \geq 4) = 1 - P(Y \leq 3)$ was required then omitted $P(Y = 0)$ when evaluating $P(Y \leq 3)$ or used $\lambda = 7$ in their calculations. The reasons given in part (d) were not always expressed well enough to gain credit.

Q9.

There were many fully-correct solutions seen to this question. The vast majority of candidates used the tables provided to enable them to find the correct answer to part (a)(i) by the most efficient method. A minority of candidates seemed only able to use the formula and consequently worked out the individual probabilities before obtaining their answer. In part (a)(ii), the formula was often used correctly to obtain the correct answer of 0.0471. However, there were some candidates who seemed to think that they had to use tables whatever the value of λ and so attempted to average the values found in the tables under $\lambda = 1.4$ and $\lambda = 1.6$ incorrectly thinking that this would give them the required value for $\lambda = 1.5$.

In part (b)(i), most candidates interpreted ‘more than 5’ correctly using $P(X + Y > 5) = 1 - P(X + Y \leq 5)$, whilst others incorrectly used $1 - P(X + Y \leq 4)$ or $1 - P(X + Y \leq 6)$. Only the most able candidates then went on to use their answer to correctly determine the numerical solution to part (b)(ii). There were many erroneous attempts at using various expressions with $0.631^7 + 0.631^8$ or $0.631^7 \times 0.369$ or even $6.5 \times 0.631^7 \times 0.369$ seen, rather than the required $p = {}^8C_7(0.631)^7(0.369) + (0.631)^8$. In part (c)(i), the vast majority of candidates correctly stated that $\bar{x} = 8$ and that $s^2 = 16.9$, but with some stating the sample variance as 15.2. These answers usually enabled candidates to indicate correctly in part (c)(ii) that a Poisson distribution was probably not a good model since the mean and the variance had very different values.

Q10.

In part (a)(i), there were surprisingly many candidates who did not know the relationship between the value of λ and the standard deviation of a Poisson distribution. In part (a)(ii), although the vast majority realised that the most efficient method of answering this part of the question required the use of the table on page 23 of the formulae booklet, very many failed to look up the appropriate values to enable them to show that $P(6 < X < 12) = P(X \leq 11) - P(X \leq 6) = 0.596$. Those candidates who calculated the separate probabilities for $X = 7, 8, 9, 10$ and 11 often reached the required answer.

In part (b)(i), it should be noted that $\lambda = 11.5$ and an indication that the distribution of T is Poisson were both required to gain the mark. In part (b)(ii), most candidates realised that “fewer than 2” implied that $P(T \leq 1) = P(T = 0) + P(T = 1)$. The correct answer of 0.000127

was then usually found by the use of $\lambda = 11.5$ and the formula $P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$ (found on page 11 of the formulae booklet).

Although part (c) asked candidates to calculate appropriate numerical measures (plural), this was ignored by some candidates. Such candidates usually calculated $\bar{x}$ and, finding this to coincide with the given value of λ , stated inappropriately that Po(12) was a suitable model for the given data. Of those candidates who did calculate more than one numerical measure, most correctly calculated the mean and variance of the given data ($\bar{x} = 12$; $s^2 = 19.3$ or $\sigma^2 = 17.4$), and then indicated that, because the mean and variance were (very) different values, Po(12) was not a suitable model for the number of places booked each week. Some candidates incorrectly thought that the mean and the standard deviation had to be the same.

Q11.

Part (a) was generally answered well. However, there were some candidates who attempted to use tables in part (a)(i), even though values are not tabulated for $\lambda = 3.5$, whilst in part (a)(iii) some candidates could not interpret ‘fewer than 10’ to mean ‘ ≤ 9 ’. It was, however, very pleasing to see fewer candidates than usual attempting to use the formula in part (a)(ii). Only the most able candidates managed to complete part (b) successfully, with the majority not realising that there were 5 ways of selecting 4 days from 5.

Part (c) was done really well by most candidates, although, in part (c)(ii), several thought that the mean had to be equal to the variance in order for a Poisson distribution to form a suitable model. Here, $\bar{x} = 9.2$ and $s^2 = 9.29$ were close enough to each other to draw this conclusion.

Q12.

Part (a)(i) was answered successfully by all but the weakest candidates, with most obtaining the correct value of 0.251. Unfortunately some failed to write their answer to 3 significant figures. In part (a)(ii), too many candidates considered $Po(4.5)$ instead of evaluating $(0.251)^3$ to obtain the correct answer of 0.0158. Most candidates realised that in order to achieve credit in part (b)(i), they had to state both the distribution (Poisson) **and** the value of its parameter ($\lambda = 9.0$). There were many excellent solutions seen to part (b)(ii) with most candidates realising that to find 'P(at least 12)' implies that $P(Y \geq 12) = 1 - P(Y \leq 11)$ has to be considered. As on previous papers, candidates found it very difficult to answer the type of question posed in part (c). Many attempts did not relate to the context used in the question but simply stated general conditions under which a Poisson distribution may be appropriate.

Q13.

In part (a), the vast majority of candidates were able to state that $A = 15$, with many going on to obtain full credit for the correct answer of 0.181. Candidates still had difficulty with the interpretation of 'more than' with the result that $P(X > 18)$ or $P(X \geq 18) = 1 - P(X \leq 17)$ was often seen. In part (b)(i), several candidates used $A = 7$ instead of combining the two Poisson variables X and Y to obtain $X + Y \sim P_0(10)$.

Interpretation of 'fewer than 15' was usually stated correctly as ' < 14 ', with tables then used to obtain the required answer 0.917. In part (b)(ii), the idea of 'independence' was usually well understood although some candidates stated incorrectly that the variable was normal.

Q14.

This was generally well answered. Some confused standard deviation with variance in part (b)(i).

Q15.

Confusion between variance and standard deviation was not uncommon. Only candidates who realised they were supposed to compare the mean and variance of the given data made progress in part (b)(ii).

Q16.

The numerical parts were usually answered correctly but many candidates were unable to provide a convincing answer to part (c).

Q17.

This question was generally well answered. Some candidates in part (c)(i), instead of calculating the variance, assumed it was equal to the mean and so were unable to answer part (c)(ii).

Q18.

Using tables to find Poisson probabilities posed few problems. However, not all candidates used the appropriate mean when Poisson distributions were combined.